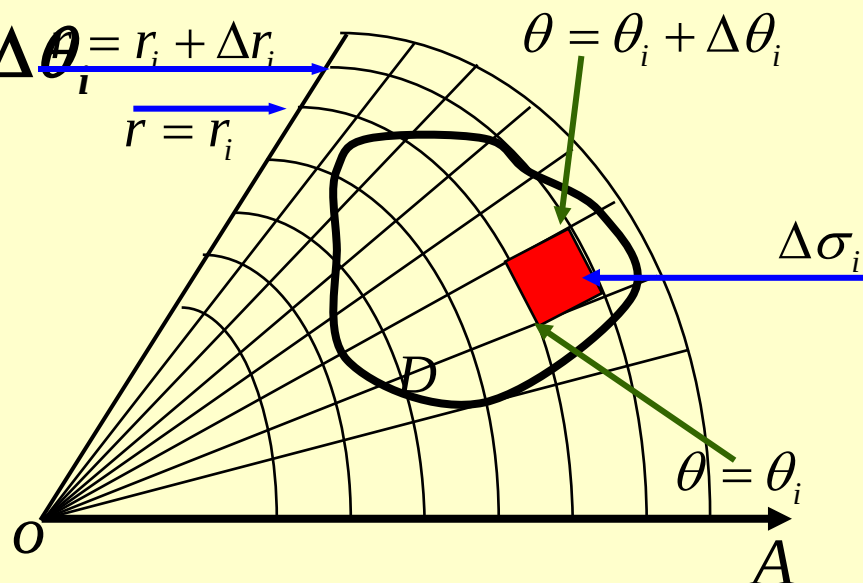


二重积分的计算法 (2)

一、利用极坐标系计算二重积分

$$\begin{aligned}\Delta\sigma_i &= \frac{1}{2}(r_i + \Delta r_i)^2 \cdot \Delta\theta_i - \frac{1}{2}r_i^2 \cdot \Delta\theta_i \\ &= \frac{1}{2}(2r_i + \Delta r_i)\Delta r_i \cdot \Delta\theta_i \\ &= \frac{r_i + (r_i + \Delta r_i)}{2} \Delta r_i \cdot \Delta\theta_i \\ &= \bar{r}_i \cdot \Delta r_i \cdot \Delta\theta_i,\end{aligned}$$



$$\iint_D f(x, y) dx dy = \iint_D f(r \cos \theta, r \sin \theta) r dr d\theta.$$

二重积分化为二次积分的公式 (1)

区域特征如图

极点在区域之外

$$\alpha \leq \theta \leq \beta,$$

$$\varphi_1(\theta) \leq r \leq \varphi_2(\theta).$$

$$\iint_D f(r \cos \theta, r \sin \theta) r dr d\theta$$

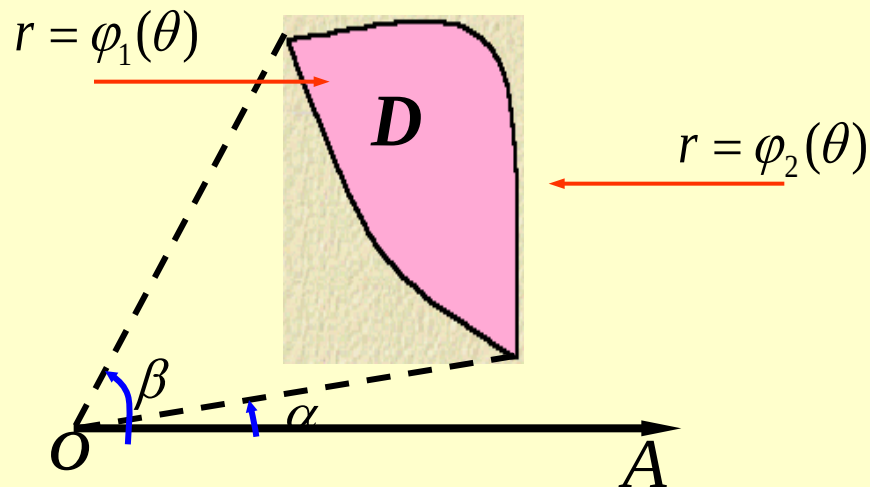
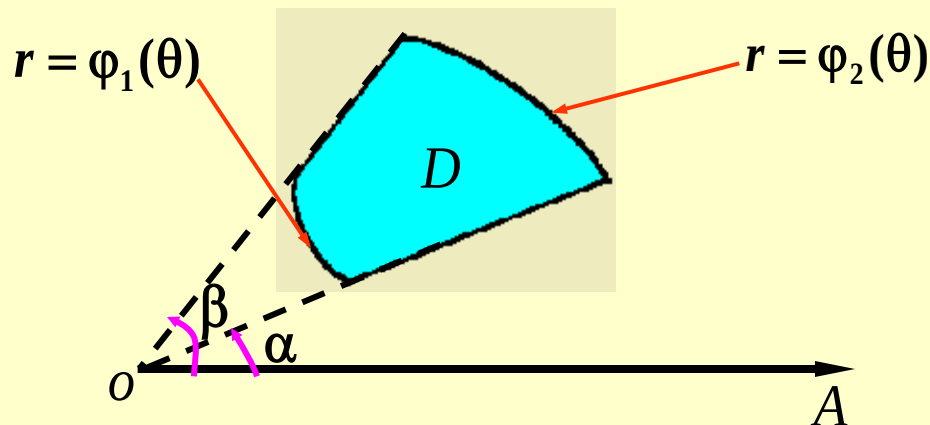
$$= \int_{\alpha}^{\beta} d\theta \int_{\varphi_1(\theta)}^{\varphi_2(\theta)} f(r \cos \theta, r \sin \theta) r dr.$$

区域特征如图

$$\alpha \leq \theta \leq \beta, \quad \varphi_1(\theta) \leq r \leq \varphi_2(\theta).$$

$$\iint_D f(r \cos \theta, r \sin \theta) r dr d\theta$$

$$= \int_{\alpha}^{\beta} d\theta \int_{\varphi_1(\theta)}^{\varphi_2(\theta)} f(r \cos \theta, r \sin \theta) r dr.$$



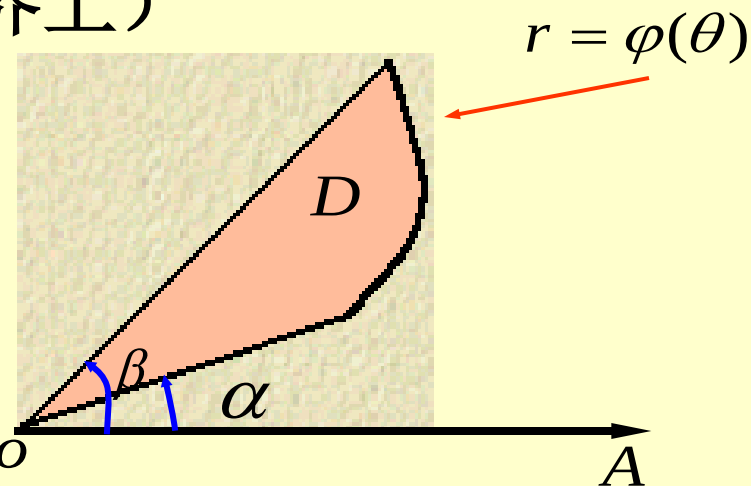
二重积分化为二次积分的公式（2）

区域特征如图（极点在D的边界上）

$$\alpha \leq \theta \leq \beta,$$

$$0 \leq r \leq \varphi(\theta).$$

$$\iint_D f(r \cos \theta, r \sin \theta) r dr d\theta$$
$$= \int_{\alpha}^{\beta} d\theta \int_0^{\varphi(\theta)} f(r \cos \theta, r \sin \theta) r dr.$$

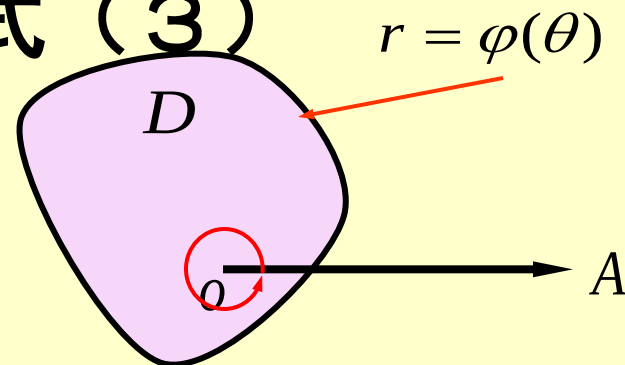


注意内下限未必全为0

二重积分化为二次积分的公式（3）

区域特征如图

（极点在D的内部）



$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq \varphi(\theta).$$

$$\begin{aligned} \iint_D f(r \cos \theta, r \sin \theta) r dr d\theta \\ = \int_0^{2\pi} d\theta \int_0^{\varphi(\theta)} f(r \cos \theta, r \sin \theta) r dr. \end{aligned}$$

极坐标系下区域的面积 $\sigma = \iint_D r dr d\theta.$

例 1 写出积分 $\iint_D f(x, y) dx dy$ 的极坐标二次积分形式, 其中积分区域

$$D = \{(x, y) \mid 1 - x \leq y \leq \sqrt{1 - x^2}, 0 \leq x \leq 1\}.$$

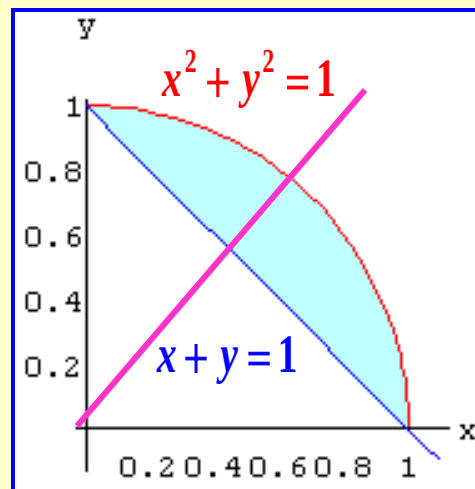
解 在极坐标系下
$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

所以圆方程为 $r = 1$,

直线方程为 $r = \frac{1}{\sin \theta + \cos \theta}$,

$$\iint_D f(x, y) dx dy$$

$$= \int_0^{\frac{\pi}{2}} d\theta \int_{\frac{1}{\sin \theta + \cos \theta}}^1 f(r \cos \theta, r \sin \theta) r dr.$$



例2 计算 $\iint_D e^{-x^2-y^2} dx dy$, 其中 D 是由中心在
原点, 半径为 a 的圆周所围成的闭区域.

解 在极坐标系下

$$D: 0 \leq r \leq a, 0 \leq \theta \leq 2\pi.$$

$$\iint_D e^{-x^2-y^2} dx dy = \int_0^{2\pi} d\theta \int_0^a e^{-r^2} r dr = \pi(1 - e^{-a^2}).$$

例 3 求广义积分 $\int_0^\infty e^{-x^2} dx$.

解 $D_1 = \{(x, y) \mid x^2 + y^2 \leq R^2\}$

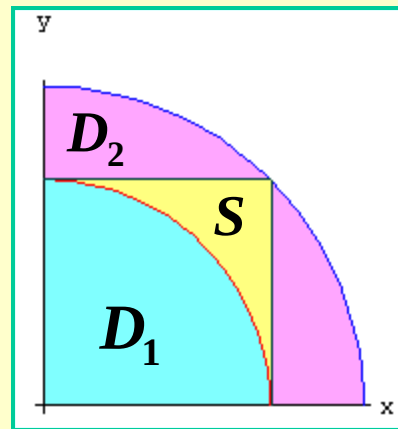
$$D_2 = \{(x, y) \mid x^2 + y^2 \leq 2R^2\}$$

$$S = \{(x, y) \mid 0 \leq x \leq R, 0 \leq y \leq R\}$$

$$\{x \geq 0, y \geq 0\} \quad \text{显然有 } D_1 \subset S \subset D_2$$

$$\because e^{-x^2-y^2} > 0,$$

$$\therefore \iint_{D_1} e^{-x^2-y^2} dx dy \leq \iint_S e^{-x^2-y^2} dx dy \leq \iint_{D_2} e^{-x^2-y^2} dx dy.$$



$$\text{又} \because I = \iint_S e^{-x^2-y^2} dx dy$$

$$= \int_0^R e^{-x^2} dx \int_0^R e^{-y^2} dy = \left(\int_0^R e^{-x^2} dx \right)^2;$$

$$I_1 = \iint_{D_1} e^{-x^2-y^2} dx dy = \int_0^{\frac{\pi}{2}} d\theta \int_0^R e^{-r^2} r dr = \frac{\pi}{4} (1 - e^{-R^2});$$

$$\text{同理 } I_2 = \iint_{D_2} e^{-x^2-y^2} dx dy = \frac{\pi}{4} (1 - e^{-2R^2});$$

$$\because I_1 < I < I_2,$$

$$\therefore \frac{\pi}{4} (1 - e^{-R^2}) < \left(\int_0^R e^{-x^2} dx \right)^2 < \frac{\pi}{4} (1 - e^{-2R^2});$$

$$\text{当 } R \rightarrow \infty \text{ 时, } \quad I_1 \rightarrow \frac{\pi}{4}, \quad I_2 \rightarrow \frac{\pi}{4},$$

故当 $R \rightarrow \infty$ 时, $I \rightarrow \frac{\pi}{4}$, 即 $(\int_0^\infty e^{-x^2} dx)^2 = \frac{\pi}{4}$,

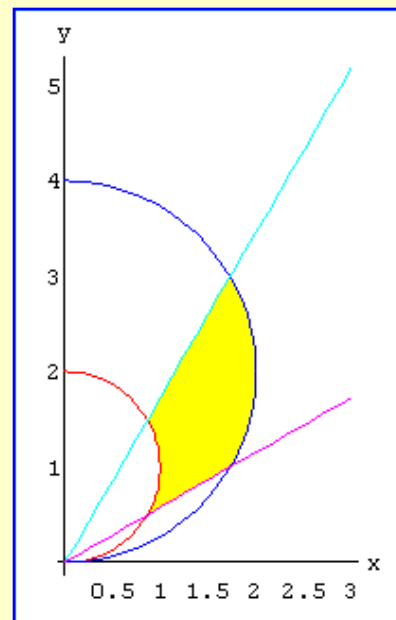
所求广义积分 $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$.

例 4 计算 $\iint_D (x^2 + y^2) dx dy$, 其 D 为由圆

$x^2 + y^2 = 2y$, $x^2 + y^2 = 4y$ 及直线 $x - \sqrt{3}y = 0$,
 $y - \sqrt{3}x = 0$ 所围成的平面闭区域.

解

$$\begin{aligned} y - \sqrt{3}x &= 0 & \Rightarrow \theta_2 &= \frac{\pi}{3} \\ x^2 + y^2 &= 4y & \Rightarrow r &= 4\sin\theta \\ x - \sqrt{3}y &= 0 & \Rightarrow \theta_1 &= \frac{\pi}{6} \end{aligned}$$



$$x^2 + y^2 = 2y \quad \Rightarrow r = 2\sin\theta$$

$$\iint_D (x^2 + y^2) dx dy = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} d\theta \int_{2\sin\theta}^{4\sin\theta} r^2 \cdot r dr = 15\left(\frac{\pi}{2} - \sqrt{3}\right).$$

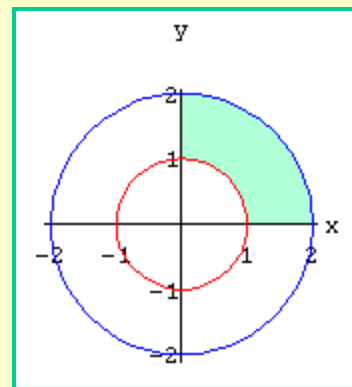
例 5 计算二重积分 $\iint_D \frac{\sin(\pi\sqrt{x^2 + y^2})}{\sqrt{x^2 + y^2}} dx dy$,

其中积分区域为 $D = \{(x, y) | 1 \leq x^2 + y^2 \leq 4\}$.

解 由对称性, 可只考虑第一象限部分,

$$D = 4D_1$$

注意: 被积函数也要有对称性.



$$\begin{aligned}\iint_D \frac{\sin(\pi\sqrt{x^2+y^2})}{\sqrt{x^2+y^2}} dx dy &= 4 \iint_{D_1} \frac{\sin(\pi\sqrt{x^2+y^2})}{\sqrt{x^2+y^2}} dx dy \\ &= 4 \int_0^{\frac{\pi}{2}} d\theta \int_1^2 \frac{\sin \pi r}{r} r dr = -4.\end{aligned}$$

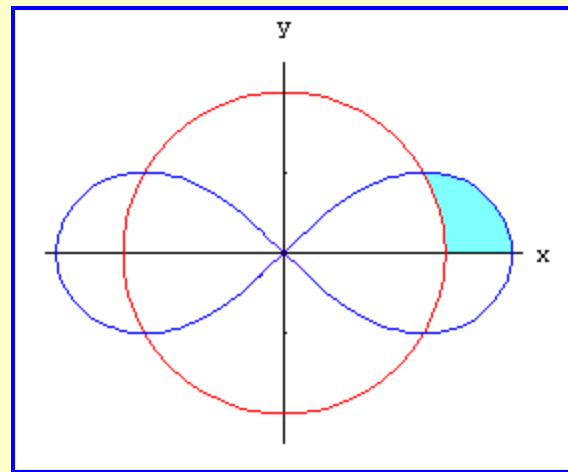
例 6 求曲线 $(x^2 + y^2)^2 = 2a^2(x^2 - y^2)$
和 $x^2 + y^2 \geq a^2$ 所围成的图形的面积.

解 根据对称性有 $D = 4D_1$

在极坐标系下

$$x^2 + y^2 = a^2 \Rightarrow r = a,$$

$$(x^2 + y^2)^2 = 2a^2(x^2 - y^2) \Rightarrow r = a\sqrt{2\cos 2\theta},$$



$$\text{由} \begin{cases} r = a\sqrt{2\cos 2\theta} \\ r = a \end{cases}, \quad \text{得交点 } A = (a, \frac{\pi}{6}),$$

$$\begin{aligned} \text{所求面积 } \sigma &= \iint_D dx dy = 4 \iint_{D_1} dx dy \\ &= 4 \int_0^{\frac{\pi}{6}} d\theta \int_a^{a\sqrt{2\cos 2\theta}} r dr = a^2 \left(\sqrt{3} - \frac{\pi}{3} \right). \end{aligned}$$

例7 计算 $\iint_D \sqrt{R^2 - x^2 - y^2} d\sigma, D: x^2 + y^2 \leq Rx$

解

$$\begin{aligned} I &= \int_{-\pi/2}^{\pi/2} d\theta \int_0^{R\cos\theta} \sqrt{R^2 - r^2} r dr \\ &= -\frac{1}{3} \int_{-\pi/2}^{\pi/2} (R^2 - r^2)^{\frac{3}{2}} \Big|_0^{R\cos\theta} d\theta \end{aligned}$$

$$= \frac{1}{3} \int_{-\pi/2}^{\pi/2} [R^3 - (R^2 - R^2 \cos^2 \theta)^{\frac{3}{2}}] d\theta$$

$$= \frac{R^3}{3} \int_{-\pi/2}^{\pi/2} [1 - |\sin^3 \theta|] d\theta \quad (\text{注意} (\sin^2 \theta)^{\frac{3}{2}} = |\sin^3 \theta|)$$

$$= \frac{2R^3}{3} \int_0^{\pi/2} (1 - \sin^3 \theta) d\theta$$

$$= \frac{R^3}{3} \left(\pi - \frac{4}{3} \right)$$

二、小结

二重积分在极坐标下的计算公式

$$\iint_D f(r \cos \theta, r \sin \theta) r dr d\theta \begin{cases} = \int_{\alpha}^{\beta} d\theta \int_{\varphi_1(\theta)}^{\varphi_2(\theta)} f(r \cos \theta, r \sin \theta) r dr. \\ = \int_{\alpha}^{\beta} d\theta \int_0^{\varphi(\theta)} f(r \cos \theta, r \sin \theta) r dr. \\ = \int_0^{2\pi} d\theta \int_0^{\varphi(\theta)} f(r \cos \theta, r \sin \theta) r dr. \end{cases}$$

(在积分中注意使用对称性)

思考题

交换积分次序:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{a \cos \theta} f(r, \theta) dr \quad (a \geq 0).$$

思考题解答

$$D: \begin{cases} -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \\ 0 \leq r \leq a \cos \theta \end{cases},$$

$$I = \int_0^a dr \int_{-\arccos \frac{r}{a}}^{\arccos \frac{r}{a}} f(r, \theta) d\theta.$$

